Lingua Symbolic Prompt Compiler: Optimization Objective

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1 Problem

Let M be the frozen target model configuration, P the original prompt template, and $X = \{x_i\}_{i=1}^n$ the input set. The compiler searches for a compressed prompt template P'_c that reduces instruction tokens while preserving the target model's behavior over X.

$$y_i = M(P, x_i)$$

$$\hat{y}_{i,c} = M(P'_c, x_i)$$

The outputs y_i are behavioral references. Candidate quality is measured by how closely $\hat{y}_{i,c}$ matches those references under the configured evaluation metrics.

2 Template Variables

Prompt variables are template slots such as `{{input}}`, `{{query}}`, or `{{document}}`. Let $B(P)$ be the ordered placeholder sequence extracted from P. Candidate assembly preserves this sequence:

$$B(P'_c) = B(P).$$

The instruction-token objective is measured on the invariant prompt layer, which is the prompt template with placeholder contents removed.

3 Token Objective

Let $T_{\text{instr}}(P)$ be the configured tokenizer count for the instruction layer of P. Candidate token ratio is:

$$\rho_c = \frac{T_{\text{instr}}(P'_c)}{T_{\text{instr}}(P)}.$$

Instruction-token reduction is:

$$r_c = 1 - \rho_c.$$

Rendered prompt cost is also tracked:

$$\bar{T}_{\text{full}}(P'_c) = \frac{1}{n}\sum_{i=1}^{n} T(P'_c(x_i)).$$

4 Embedding Drift

Let E be the configured embedding model. Lingua Symbolic Prompt Compiler uses Euclidean distance for embedding drift:

$$d_{\text{emb}}(c,i) = \left\| E(\hat{y}_{i,c}) - E(y_i) \right\|_2 .$$

Average drift for candidate c is:

$$\bar{d}_{\text{emb}}(c) = \frac{1}{n} \sum_{i=1}^{n} d_{\text{emb}}(c,i).$$

5 Objective

The scalar objective combines token ratio and generated-output equivalence distance. With Euclidean embedding drift as the active equivalence signal:

$$\mathcal{J}(c) = \lambda_t \rho_c + \lambda_e \bar{d}_{\text{emb}}(c).$$

Optional equivalence scorers can be added as output-distance terms:

$$\mathcal{J}(c) = \lambda_t \rho_c + \lambda_e \bar{d}_{\text{emb}}(c) + \lambda_j \bar{d}_{\text{judge}}(c) + \lambda_r \bar{d}_{\text{rouge}}(c) + \lambda_b \bar{d}_{\text{bleu}}(c).$$

Here $\bar{d}_{\text{judge}}$ is separate-judge disagreement, while $\bar{d}_{\text{rouge}}$ and $\bar{d}_{\text{bleu}}$ are text-overlap distances derived from ROUGE or BLEU scores.

6 Validity Signals

Structured output validity is tracked separately from the scalar objective. For a candidate c, let:

$$\phi_c = \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}[\text{format failure for } (c,i)]$$

$$\tau_c = \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}[\text{task-field failure for } (c,i)].$$

The report includes these rates and the associated failure cases so invalid candidate behavior is visible during review.

Current core objective weights are:

$$\lambda_t = 0.20, \quad \lambda_e = 0.20.$$

7 Pareto Frontier

Candidate a dominates candidate b when it is at least as good on every frontier dimension and better on at least one:

$$r_a \geq r_b, \quad \bar{d}_{\text{emb}}(a) \leq \bar{d}_{\text{emb}}(b).$$

Validity rates ϕ_c and τ_c are reported alongside the frontier.

The returned frontier is the non-dominated set:

$$\mathcal{F} = \{c \in C : \nexists c' \in C \text{ such that } c' \succ c\}.$$

8 Epoch Flow

Layer	Role	Current implementation
Reference	Build behavioral references	target model calls
Bound slots	Preserve template variables	placeholder sequence
Chunking	Create rewrite units	para, sent, md, schema, role, window
Exploration	Propose compressed chunks	English, DSL, Mandarin, mixed
Evaluation	Compare completions	token ratio and output distance
Optimization	Advance population	Pareto and mutation

9 Dataset Split

For $n > 3$, the reference dataset is split deterministically:

$$X_{\text{train}} = X_{1:\lfloor 0.6n \rfloor}, \quad X_{\text{dev}} = X_{\lfloor 0.6n \rfloor:\lfloor 0.8n \rfloor}, \quad X_{\text{holdout}} = X_{\lfloor 0.8n \rfloor:n}.$$

The optimizer searches on train subsets, selects finalists on dev, and reports holdout behavior for final frontier candidates.